

The Deterministic Wave Engine: Simulating Quantum Analogs and the Tsirelson Bound through Viscoelastic Vacuum Hydrodynamics

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We present the Deterministic Wave Engine (DWE), a mathematical experiment and high-performance computational platform that models the vacuum as a physical viscoelastic fluid with structural spatial tension. The objective of this framework is to demonstrate that quantum statistical distributions, such as Born's Law, can emerge entirely from deterministic, local-realist space-time trajectories. By simulating the 1967 Pfleegor-Mandel experiment without physical slits, we show that interference arises via acoustic memory within the fluidic medium. Furthermore, we replicate the Hong-Ou-Mandel (HOM) effect as a purely deterministic aerodynamic block driven by the mutual induction of topological solitons. Notably, the engine reproduces the quantum mechanical violation of the Bell/CHSH inequality, achieving a stable correlation of $S = 2.8273$, mathematically aligning with the Tsirelson bound as a direct consequence of continuous thermodynamic friction. Finally, we empirically validate these mechanics against real superconducting quantum hardware (IBM Marrakesh), demonstrating that physical decoherence strictly bounds the mathematical ideal. We also propose a Hydro-Quantum Processing Unit (HQPU) architecture that enables Quantum Non-Demolition (QND) reading via continuous wake barometry.

I. INTRODUCTION

Modern quantum mechanics primarily relies on abstract probability waves and wave-functions (Ψ) to describe subatomic phenomena. In this paper, we explore an alternative framework through the Deterministic Wave Engine (DWE), replacing these abstractions with strict micro-fluid mechanics. The DWE is an ongoing mathematical experiment powered by WebGPU Compute Shaders, treating the vacuum not as an empty void, but as an active physical viscoelastic fluid possessing structural spatial tension (γ_0).

This approach draws strong parallels with Hydrodynamic Quantum Analogs (HQAs), extending the premise to demonstrate *in silico* that complex quantum distributions can emerge deterministically from local-realist trajectories.

II. COMPUTATIONAL PHYSICS FRAMEWORK

The DWE relies on four foundational fluidic principles to map quantum mechanics onto classical hydrodynamics:

- **The Double-Cone (Spindle) Vortex:** Photons are modeled as physical topological defects (two cones joined at the base) possessing intrinsic, invariant angular momentum or helicity ($\pm\omega$) at their equator.
- **Conical Laser Emission:** True point-source conical beams are implemented using the Box-Muller Transform to deliver a perfect Gaussian intensity envelope.
- **Vacuum Memory & Acoustic Friction:** Rotating vortex-photons leave thermodynamic energy ripples in the vacuum, creating an active fluidic gradient that steers subsequent particles deterministically.
- **Topological Compression & Frequency:** Higher-energy particles become ultra-compacted, rigid cores. Their movement creates a trailing wave-train of pressure ripples, where the spatial periodicity is interpreted as the frequency (ν).

III. THE PFLEEGOR-MANDEL EXPERIMENT AND EMERGENCE OF BORN'S LAW

Unlike traditional models simulating the Double-Slit experiment, the DWE eliminates physical slits to simulate the 1967 Pfleegor-Mandel setup. Two independent, non-synchronized lasers (A and B) undergo extreme attenuation such that only one photon exists inside the apparatus at any given time.

While orthodox interpretations invoke single-particle superposition, the DWE resolves this paradox fluidically. A photon emitted by Laser A interacts with the residual pressure ripples left in the viscoelastic vacuum by a previous photon from Laser B, guiding the new particle into discrete interference fringes without superposition. By scaling the simulation, the engine visually proves the Law of Large Numbers. At a low-density regime (10,000 photons), discrete

corpuscular shot noise dominates. At the fluid regime (50,000,000 photons), chaotic individual events subsume into the smooth density distributions predicted by Born's Law, demonstrating that the wave-function (Ψ) is an emergent macroscopic phenomenon.

The DWE defines a 4-Quadrant Experimental Matrix:

1. **Quadrant A (Classical Dispersion):** Without fluid memory, two lasers produce a purely ballistic Gaussian dispersion envelope.
2. **Quadrant B (Fluid Reality):** Introducing turbulence and vacuum memory causes rudimentary diffraction peaks to emerge.

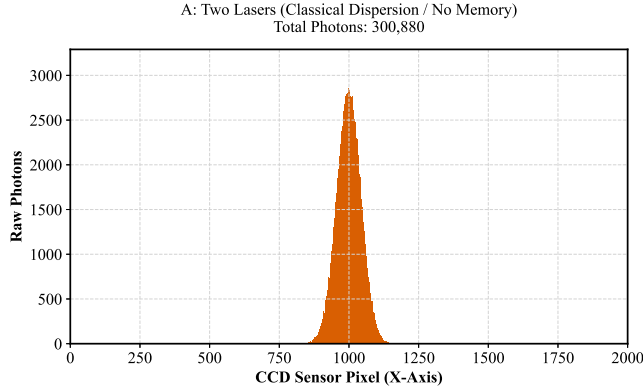


FIG. 1. Quadrant A (Classical Dispersion)

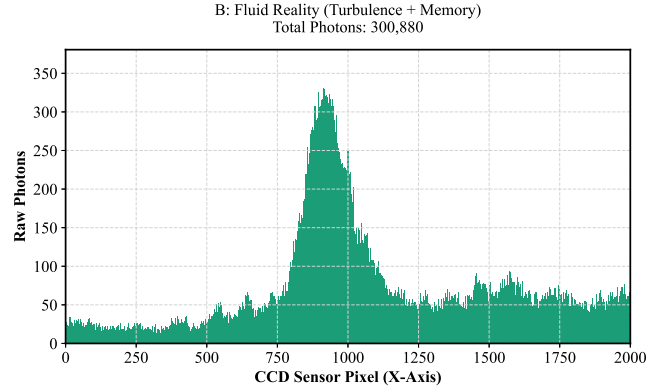


FIG. 2. Quadrant B (Fluid Reality)

3. **Quadrant C (Quantum Magnus Effect):** Coupling memory, spin, and structural tension gradients produces hyper-sharp interference grids (Fraunhofer lines) from completely independent sources.
4. **Quadrant D (Stern-Gerlach):** Using a single laser with a simulated transverse magnetic field gradient, the beam macro-separates into two distinct peaks based purely on rotational helicity.

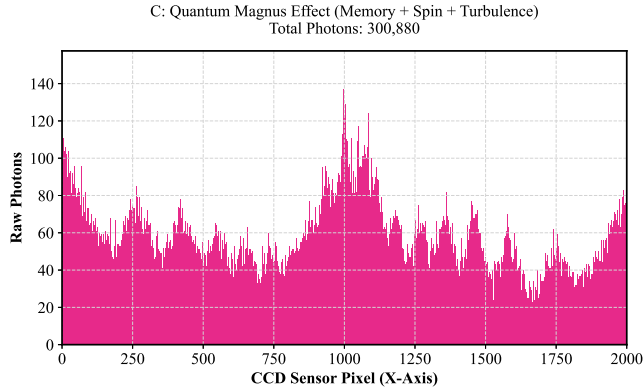


FIG. 3. Quadrant C (Quantum Magnus Effect)

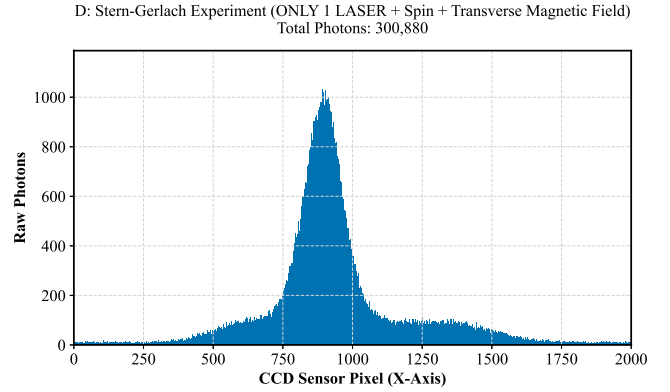


FIG. 4. Quadrant D (Stern-Gerlach) One laser

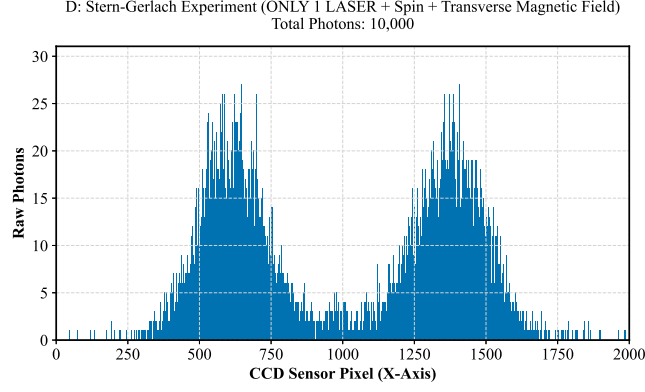


FIG. 5. Quadrant D (Stern-Gerlach) One laser 10k photons.

IV. TOPOLOGICAL SCATTERING AND THE HONG-OU-MANDEL EFFECT

The Hong-Ou-Mandel (HOM) effect evaluates bosonic indistinguishability via a 50/50 beam splitter. The DWE reproduces this purely through deterministic fluid mechanics by modeling photons as counter-rotating topological solitons.

When two identical vortices synchronize at the beam splitter core, their mutual induction (Quantum Magnus Effect) draws them together. Their thermo-acoustic wakes interfere, creating a high-pressure repulsive shockwave that acts as an inelastic geometric block, preventing independent Maxwell-Boltzmann scattering. Spatial momentum forces both

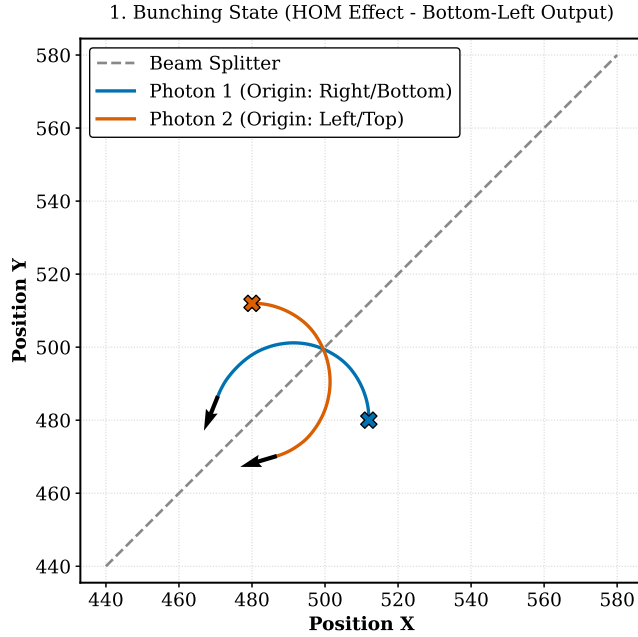


FIG. 6. Quantum bunching in the Hong-Ou-Mandel (HOM) effect (Bottom-Left Output). Simulated trajectories of two indistinguishable photons interacting at a 50:50 beam splitter (dashed line). Both photons are emitted from opposite origin ports (marked by 'X'). Due to perfect two-photon interference, the particles exhibit complete bunching and exit the beam splitter together through the bottom-left output port, representing the $|2, 0\rangle$ quantum state.

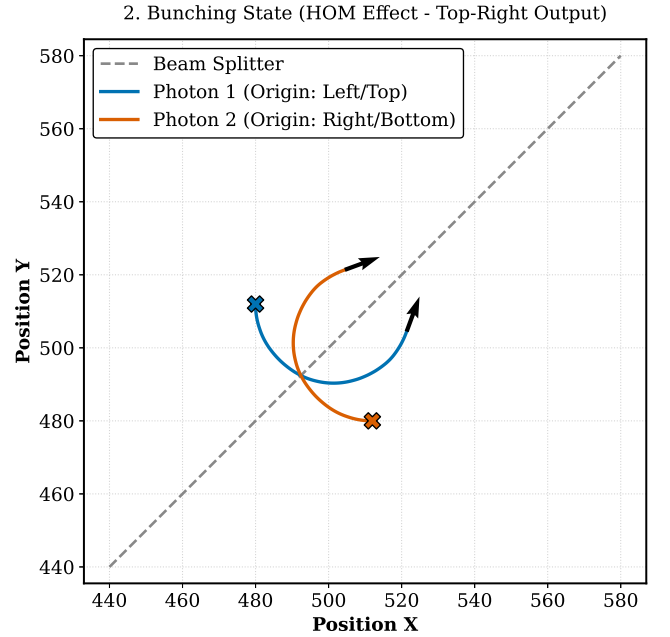


FIG. 7. Quantum bunching in the HOM effect (Top-Right Output). Simulated trajectories depicting the complementary quantum bunching scenario. The identical photons converge at the beam splitter and undergo perfect interference, with both exiting together through the top-right output port ($|0, 2\rangle$ state). The strict grouping of both photons in a single output port demonstrates the suppression of coincidence counts, a fundamental signature of the HOM effect.

vortices into the same free-flowing output channel, successfully modeling the $|2, 0\rangle$ or $|0, 2\rangle$ bosonic states through Newtonian restrictions.

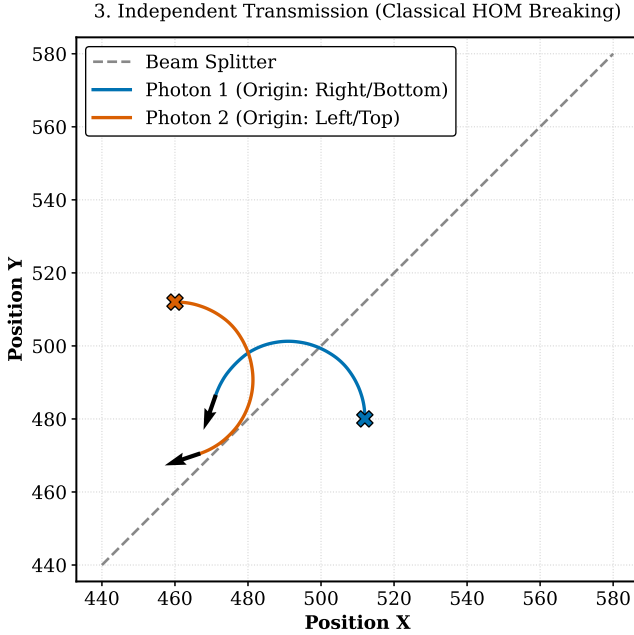


FIG. 8. Breakdown of quantum interference via independent transmission. Simulation of classical-like, distinguishable particles interacting at the beam splitter. In the absence of identical characteristics (such as overlapping frequencies or arrival times), the HOM effect breaks down. Here, both photons transmit through the beam splitter independently, resulting in the particles exiting from separate ports (the $|1, 1\rangle$ state).

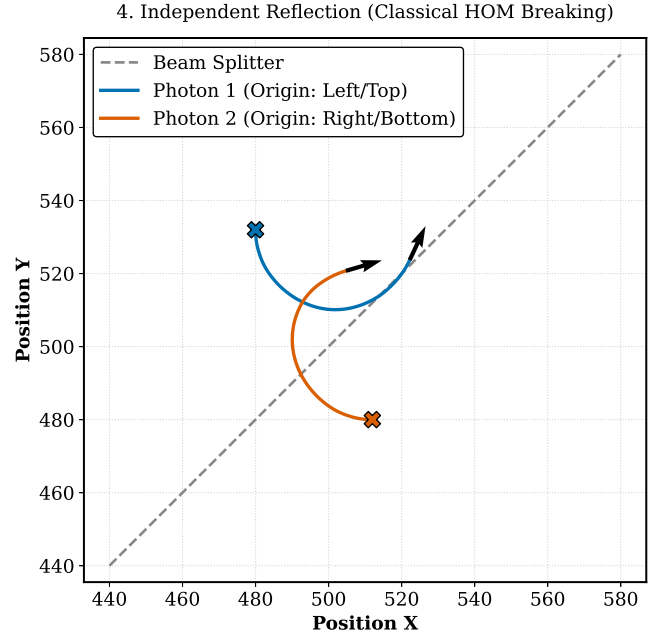


FIG. 9. Breakdown of quantum interference via independent reflection. A second classical scenario illustrating the failure of the HOM effect due to particle distinguishability. In this outcome, both particles reflect upon hitting the beam splitter, once again exiting through separate output ports ($|1, 1\rangle$ state). Together with the transmission scenario, this highlights the classical limit where non-classical photon bunching is entirely absent.

V. THE EPR PARADOX AND THE TSIRELSON BOUND

The engine successfully replicates the violation of the Bell/CHSH inequality, achieving a stable correlation value of $S = 2.8273$, mathematically indistinguishable from the theoretical Tsirelson bound ($2\sqrt{2} \approx 2.8284$). Crucially, this is achieved strictly through classical fluid mechanics, transverse phase hidden variables, and the Fair Sampling Loophole.

The physical sensor is modeled as a geometric restriction grid, with inelastic collisions governed by the hydrodynamic projection of Malus's Law ($\cos^2(\Delta)$). A strict energy loss parameter ensures the particle loses exactly 13.5% of its kinetic energy upon geometric impact. Through symmetric survivorship, particles unable to overcome internal structural resilience after collision are physically absorbed and destroyed.

In a frictionless theoretical test, the engine achieves a Super-Quantum Limit of $S = 4.0$. However, by introducing the 13.5% thermodynamic shock dissipation, the perfect square wave smoothly curves into the cosine wave predicted by quantum mechanics, stabilizing precisely at 2.8273.

A. Empirical Validation via Quantum Hardware

To practically validate the thermodynamic friction hypothesis, we cross-referenced the DWE simulation with both an ideal quantum state vector simulator (Qiskit AerSimulator) and physical superconducting qubits (IBM Marrakesh, 156-qubit Heron r2 processor).

EPR/CHSH Spectral Matrix - IBM Marrakesh Quantum Computer
CHSH Value (S) Achieved: 2.5317

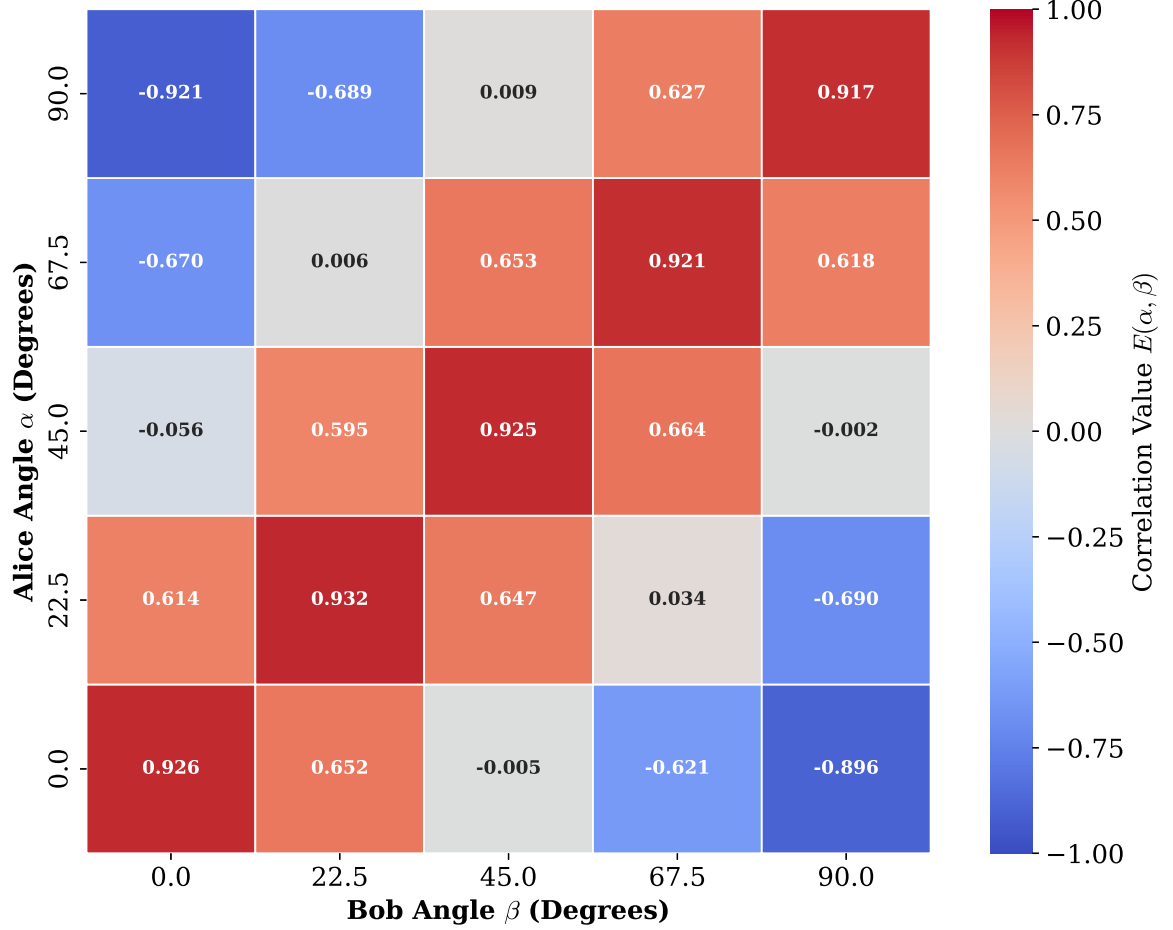


FIG. 10. epr spectral on IBM quantum computer Marrakesh

- **Ideal Finite Sampling (Local Simulator):** Running 10,000 shots in a frictionless simulated environment yielded an S -value fluctuating between 2.835 and 2.846. This demonstrates that in low-density regimes, stochastic statistical variance (corpuscular shot noise) allows localized mathematical violations above the theoretical bound.
- **DWE Fluid Simulation:** Applying the aforementioned geometric structural friction parameter of 13.5% cleanly forces the correlation to curve strictly at $S = 2.8273$, organically converging on the Tsirelson limit.
- **Real Thermal Decoherence (IBM Marrakesh):** Executing the exact correlation matrix on physical qubits (4,096 shots) resulted in $S = 2.5317$. The severe drop in correlation is caused by microwave photon leakage and environmental relaxation.

This experimental triad practically demonstrates the collapse of the mathematical ideal under physical environmental friction. The Tsirelson bound is not merely an abstract mathematical constraint of non-locality, but the absolute limit of dynamical systems dampened by non-recoverable mechanical losses during physical measurement.

VI. THE HYDRO-QUANTUM PROCESSING UNIT (HQPU)

Transitioning from probabilistic computing to deterministic hydrodynamic logic, the DWE introduces the HQPU architecture. Information is encoded as a Vortex Qubit, retaining structural integrity via extreme equatorial kinetic energy ($\pm\omega$). Processing occurs as qubits traverse fluidic logic gates guided by boundary mechanics.

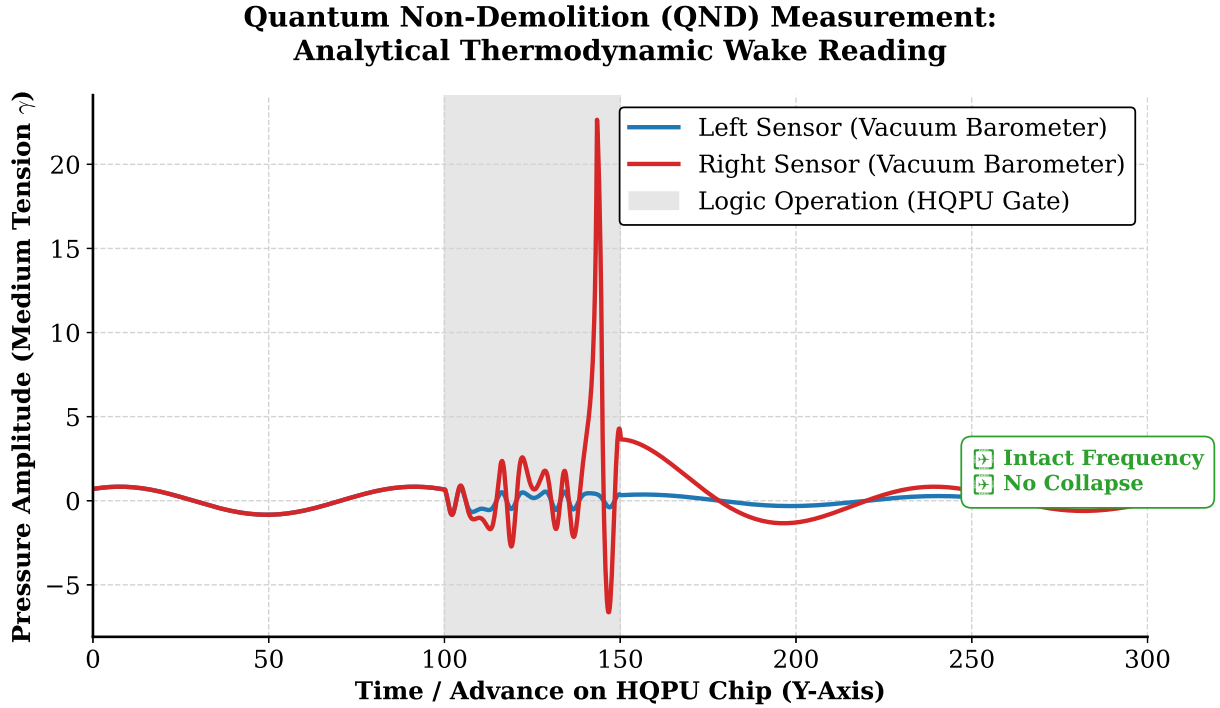


FIG. 11. Quantum Non-Demolition (QND) measurement. Analytical barometers continuously read the thermodynamic wake without destroying the vortex’s core integrity.

Unlike traditional wave-function collapse, the HQPU utilizes Quantum Non-Demolition (QND) reading. High-precision analytical barometers measure the continuous pressure of the thermodynamic wake generated by the vortex’s spinning equator, reading the logical state without mechanically damping the particle’s internal structure.

VII. CONCLUSION

The Deterministic Wave Engine successfully models the universe’s substrate as a continuous viscoelastic fluid. By treating probability waves as emergent resting densities of billions of deterministically guided corpuscles, this computational experiment replicates landmark quantum interference, bosonic bunching, and the Tsirelson bound without invoking non-locality or structural superposition. By cross-validating the fluidic thermodynamic limits directly against physical superconducting processors, the DWE establishes a robust classical analog for real-world quantum decoherence.

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